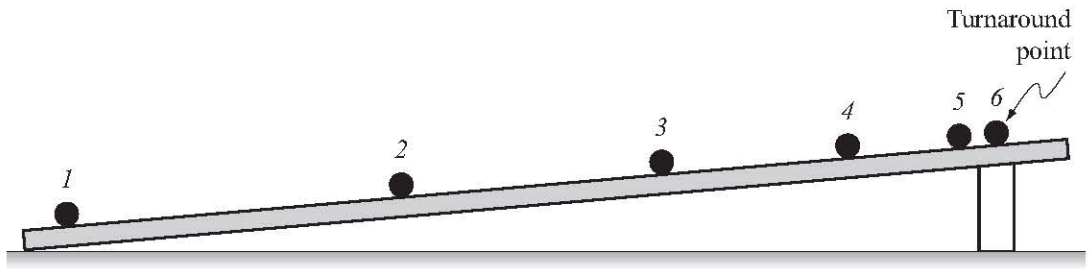
ACCELERATION IN ONE DIMENSION

I. Motion with decreasing speed

The diagram below represents a strobe photograph of a ball as it rolls *up* a track. (In a strobe photograph, the position of an object is shown at instants separated by *equal time intervals*.) The ball is already in motion at instant 1.



A. Draw vectors on your diagram that represent the instantaneous velocity of the ball at each of the instants shown. If the magnitude of the velocity is zero at any instant, indicate that explicitly. Explain why you drew the vectors as you did.

We will call diagrams like the one you drew above *velocity diagrams*. Unless otherwise specified, a velocity diagram shows both the location and the velocity of an object at instants in time that are separated by equal time intervals.

B. In the space at right, sketch vectors to represent the velocities at instants 1 and 2.

In the space at right, draw a diagram that shows how to find the vector that must be *added* to the velocity at the earlier time to equal the velocity at the later time. Label these vectors $\vec{v}_1$, $\vec{v}_2$, and $\Delta\vec{v}$.

$\vec{v}_1$:

$\vec{v}_2$:

$\Delta\vec{v}$

Why is the name *change-in-velocity* appropriate for this vector?

How does the direction of the *change-in-velocity* vector compare to the direction of the velocity vectors?

Would your answer about the directions differ if you were to select two *different* consecutive instants *(e.g.*, instants *3* and *4)* while the ball was slowing down? Explain.

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